

Sequential extraction of carotenoids and polyphenols from *Ascophyllum nodosum* using pH-responsive switchable deep eutectic solvents

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ARTICLE INFO

Keywords:

Biorefinery
Switchable deep eutectic solvents
Carotenoids
Polyphenols
Extraction efficiency
Ascophyllum nodosum
Brown algae

ABSTRACT

A sustainable approach was developed for the sequential extraction of carotenoids and polyphenols from *Ascophyllum nodosum* using a fatty acid-based switchable deep eutectic solvent (SDES) composed of octanoic (C8) and decanoic (C10) fatty acids. The polarity of the SDES was adjusted during the process, enabling pH-induced two-layer separation and simultaneous recovery of extracts. Under optimized conditions, the highest carotenoid yield was 1.25 ± 0.02 mg β -carotene equivalents/g dried seaweed (solvent-to-solid ratio 10.25 mL/g, C10:C8 molar ratio 2:1, extraction temperature of 30 °C, and extraction time of 90 min), while polyphenol extraction from the residual biomass yielded 3.97 ± 0.07 mg gallic acid equivalents (GAE)/g (40 mL/g, extraction temperature of 40 °C, and extraction time of 70 min). These yields were 81.16% and 56.30% higher than those obtained using conventional solvents (hexane and 70% ethanol), respectively. Repeated extraction experiments showed that the total extractable carotenoid content (TCC) was 2.00 mg/g dried seaweed, with approximately 81% recovered within the first two extraction cycles. The total extractable polyphenol content (TPC) was 10.04 mg GAE/g, of which 70.5% remained in the residue after carotenoid extraction. Under optimized conditions, a single extraction recovered approximately 56% of the extractable polyphenols in the residue (~39% of the total TPC), while only ~2.5% remained associated with the SDES-rich layer. The SDES demonstrated effective recyclability, with a recovery yield of $93.70 \pm 0.84\%$ and 95.20% retention of carotenoid extraction efficiency after reuse. These results demonstrate that the SDES selectively extracts carotenoids while preserving polyphenols for subsequent recovery, supporting its application in integrated seaweed biorefineries. The reduced DPPH radical scavenging activity observed in SDES is likely influenced by solvent-dependent effects, warranting further study on activity recovery after solvent exchange or formulation.

1. Introduction

Ascophyllum nodosum is a marine macroalgae or brown seaweed that is a rich source of bioactive compounds (Karatzia et al., 2012). These include polysaccharides, carotenoids and polyphenols – compounds that can exhibit antioxidant, antidiabetic (Olsthoorn et al., 2021), anticancer, anticoagulant (Zhang et al., 2022c; Zhang et al., 2022c) and anti-inflammatory activities (Remya et al., 2022). Bioactive extracts from natural sources are currently in high demand for use as novel ingredients in functional foods and other consumer products such as cosmetics (Salas et al., 2023; Xie et al., 2024). Marine macroalgae are considered a renewable and sustainable resource in comparison to other natural sources that require freshwater or arable land for cultivation, as they typically have a higher growth rate and more rapid biomass

production and require fewer resources such as synthetic fertilizers and pesticides (Pereira & Cotas, 2024). Thus, *A. nodosum* is attractive as a renewable, sustainable source of bioactive compounds.

Traditional organic solvents used to extract compounds from seaweeds are commonly highly toxic, flammable, and not environmentally friendly. For example, phenolic compounds are commonly extracted with acetone or ethanol (Cotas et al., 2020), while carotenoids are generally extracted using hexane, methanol, or acetone (Saini & Keum, 2018). In addition, conventional organic solvents typically extract either hydrophobic or hydrophilic compounds, and the remaining bioactive components in the residues are discarded, causing a considerable loss of resources and environmental pollution (Cao et al., 2018). A more sustainable way to extract bioactive compounds from seaweeds uses a biorefinery approach, which not only produces biofuel but also enables

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<https://doi.org/10.1016/j.afres.2026.102392>

Received 23 December 2025; Received in revised form 26 June 2026; Accepted 13 July 2026

Available online 14 July 2026

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the recovery of multiple bioactive co-products, creating several valuable product streams while minimizing waste (Bordoloi & Goosen, 2020).

Switchable Deep Eutectic Solvents (SDESs) are types of Deep Eutectic Solvents (DESs) or Natural Deep Eutectic Solvents (NADESs) that have generated much interest in recent years as more environmentally benign alternatives to traditional organic solvents (Hansen et al., 2021; Kalyniukova et al., 2021). They are comprised of a Hydrogen Bond Acceptor (HBA) and Hydrogen Bond Donor (HBD), resulting in a eutectic mixture with depressed melting point (Freitas et al., 2024). SDESs can be categorized according to their HBA which can include: organic ammonium halides, organic ammonium, fatty acids, alcohol amines, and alcohols (Zhang et al., 2022b). The most prevalent SDES components are menthol and fatty acids, especially C6–12 long-chain fatty acids. These fatty acids can be used as HBA and HBD simultaneously (Zhang et al., 2022b) and have several advantages including the easy modification of phase transition temperatures by combining fatty acids with different alkyl chain lengths as well as being considered natural, non-corrosive, affordable, and non-toxic (Florindo et al., 2018). Moreover, octanoic (C8) and decanoic (C10) fatty acids are found in natural food sources such as coconut and palm kernel oil and are used as food additives. This may be an advantage as these SDES components could give added functionality to crude bioactive SDES extracts that are incorporated into functional food or consumer products (Ji & Brooks, 2025). Studies have shown that some DESs can provide greater stability upon thermal and storage for bioactive compounds (Zannou et al., 2020), most likely due to interactions with the DES components, and this would be another advantage in having both the extracted compounds and SDES components present in the final extract.

Fatty acid-based SDESs can alternate between hydrophobic and hydrophilic phases by pH adjustment. For example, adding an alkaline solution deprotonates the fatty acids, resulting in a hydrophilic phase solvent, while adding an acidic solution protonates fatty acids, resulting in a hydrophobic phase solvent (Florindo et al., 2018; Zhang et al., 2022b). Researchers have also used excess amine solutions to de-protonate fatty acid-based SDESs and form a complex between the amine and fatty acids, creating a homogenous solution that can be split into two phases by adjusting the pH to aid in separation and recovery of extracted compounds (Sed et al., 2018). Studies have used SDES phase splitting in trace analysis and microextraction of analytes from water (Ataee et al., 2024; Ma & Row, 2021; Wu et al., 2023), milk (Li et al., 2020; Wang et al., 2020) and fruit juices (Fattahi et al., 2024; Hassanpour et al., 2023; Li et al., 2019).

SDES extraction studies have been conducted for the extraction of hydrophobic carotenoids from different natural sources such as the microalgae *Chlorella vulgaris* and *Tisochrysis lutea* (Xu et al., 2022 and Sportiello et al., 2024), and other plant material such as pumpkin (Stupar et al. (2021) and Sportiello et al. (2025) and carrots (Oh & Brooks, 2025). However, only a few studies have used SDES to extract hydrophilic compounds, for example the extraction of polyphenols from millet (Zhang et al., 2022a). Since pH adjustments can be used to change between hydrophobic and hydrophilic phases, SDESs can be used to extract hydrophobic and hydrophilic compounds using the same solvent system, which reduces waste. This adaptability can streamline the extraction process compared to traditional methods by eliminating the need to remove and replace specific solvents between sequential extractions. Moreover, SDES phase splitting allows for simultaneous separation and recovery of the desired compounds. In line with green extraction principles, minimizing the steps involved in processes like solvent extraction of natural products can lead to lower costs and more efficient energy use (Chemat et al., 2012). Despite these potential advantages, to the best of our knowledge, only one study (Oh & Brooks, 2025) has used SDESs to extract both hydrophobic (carotenoids) and hydrophilic (polyphenols) compounds from the same source material (carrots) using an optimized sequential approach.

There are currently no studies on marine macroalgae that have used SDES for extraction, as only conventional DES systems (that mostly use

choline chloride or betaine as the HBA), have been explored. For brown seaweed, the compounds that have been targeted for extraction with DESs include proteins or phenolic compounds from *Saccharina latissima* (Moldes et al., 2024 and Zeb et al., 2024), phenolic compounds, phlorotannins and photosynthetic pigments from *Sargassum filipendula* (Mello et al., 2025), phenolic compounds from *A. nodosum* and *Laminaria hyperborea* (Jesus et al., 2025) and alginate and fucoidan from *Sargassum macroalgae* (Guerrero-Higareda & Carrillo-Nieves, 2025).

To date, no studies have been reported on the extraction of carotenoids from *A. nodosum* using fatty acid-based SDES. Thus, the overall goal of this investigation was to develop a green extraction strategy for the sequential recovery of hydrophobic (carotenoids) compounds and hydrophilic (phenolic) compounds from *A. nodosum* with the same fatty acid-based SDES. The sequential approach looked at first optimizing the extraction of carotenoids followed by optimizing the extraction of polyphenols from *A. nodosum*. In addition, recycling the fatty acid-based SDES was investigated and repeated extractions with fresh SDES were also performed to determine the maximum carotenoids recovered from *A. nodosum*. As well, the relationship between the SDES solvent environment and the measured DPPH radical scavenging activity of the extracted carotenoids was examined.

2. Materials and methods

2.1. Materials and chemicals

Ascophyllum nodosum, was donated by a seaweed processing company (Nova Scotia, Canada) after the material had undergone commercial outdoor solar drying to a set moisture level and then milled to a 40 mesh size fraction. The dried seaweed powder (moisture content: $11.33 \pm 0.13\%$) was placed in sealed plastic bags and stored in a desiccator kept inside a $-20\text{ }^{\circ}\text{C}$ freezer for no >12 months before use. Standards of β -carotene (ACS, $\geq 99\%$), gallic acid (ACS, $\geq 98\%$), C8 or octanoic acid (ACS, $\geq 99\%$), C10 or decanoic acid (ACS, $\geq 99\%$), C12 or dodecanoic acid (ACS, $\geq 99\%$), and 2,2-diphenyl-1-picrylhydrazyl or DPPH (ACS, $\geq 99\%$) were purchased from Sigma-Aldrich (Oakville, Ontario, Canada). Folin-Ciocalteu reagent (2 N), sodium hydroxide (ACS, 99–100% dry basis), and hydrochloric acid (37% volume) were purchased from Fisher Scientific (Ottawa, ON, Canada). Ethyl alcohol (95% volume) was obtained from Greenfield Specialty Alcohols Inc. (Toronto, Ontario, Canada). Deionized water was obtained from a Milli-Q system (Millipore, Bedford, MA, USA).

2.2. Preparation of SDESs

SDESs were synthesized with fatty acids (C8 and C10) at different molar ratios (0.5, 1.25 and 2 of C10/C8), following the SDES compositions that were studied in Stupar et al. (2021). For each hydrophobic SDES, the corresponding mixture of fatty acids was heated at $60\text{ }^{\circ}\text{C}$ in a sealed glass vial until a homogeneous clear liquid was obtained. The solvents were then slowly cooled to ambient temperature for further use (Yang et al., 2019).

2.3. Determination of total carotenoid content (TCC)

The total carotenoid content (TCC) of sample extracts was determined following the method by Nagaraj et al. (2022) with modification (Nagaraj et al., 2022). β -carotene calibration curves were constructed (from 0.5 - 30 $\mu\text{g/mL}$) using hexane or SDES, depending on the solvent used for extracting the samples. The absorbance of the β -carotene standard was measured using a UV–Vis spectrophotometer (Genesys 10S UV/Visible spectrophotometer, Thermo Scientific, USA) at 450 nm and calibrated with a blank that contained solely of hexane or SDES, as appropriate. All analyses were conducted in triplicate and the mean TCC yield was expressed as mg of β -carotene equivalent/g of original *A. nodosum* powder.

2.4. Determination of total polyphenol content (TPC)

The total polyphenol content (TPC) of sample extracts was measured employing the Folin-Ciocalteu method (Singleton et al., 1999). Briefly, 100 μ L of the sample extract solution was mixed with 200 μ L of Folin-Ciocalteu reagent (diluted with water at a volume ratio of 1:10). After 5 min of incubation, 7.5% (w/v) aqueous sodium carbonate solution (800 μ L) was added. The solution was vortexed and incubated for 2 h in the dark at room temperature (around 20 °C). The mixture was then centrifuged at 4000 rpm for 1 min. TPC was determined at the wavelength of maximum absorption (760 nm) and calibrated with a blank that contained solely water. Gallic acid was used as the standard and a gallic acid calibration curve was constructed (from 10 - 200 μ g/mL) using water as the solvent. All analyses were conducted in triplicate and the mean TPC yield was expressed as mg of gallic acid equivalent/g of dried *A. nodosum* residue.

2.5. Sequential extraction of bioactive compounds using SDESs

The sequential extraction process followed the method by Stupar et al. (2021) as a basis for extracting carotenoids using an SDES, but with additional steps to separate and analyze extracted polyphenols (Zhang et al., 2022a). The extraction process comprised two main stages: (1) optimization of total carotenoid content (TCC) and (2) optimization of total polyphenolic content (TPC), and all extractions were carried out in triplicate ($n = 3$).

The first extraction stage for carotenoids used the SDES in its hydrophobic phase and the optimization variables were the solvent/solid ratio (v/w), the molar ratio of the HBA/HBD fatty acids, and the extraction time. The HBA/HBD molar ratio was included as an optimization variable as the study by Sportiello et al. (2024), indicated that the polarity of the SDES had a significant impact on carotenoid extraction from microalgae. The extraction temperature was fixed at 30 °C to avoid carotenoid degradation (Sportiello et al., 2024). Each extraction experiment was done in a 50-mL centrifuge tube and involved adding 0.5 g of dried *A. nodosum* seaweed powder to an appropriate volume (2.5, 5 and 7.5 mL) of hydrophobic SDES at the desired solvent/solid ratio (5, 10 and 15 mL/g), where the SDES was comprised of fatty acids at the desired HBA: HBD molar ratio (0.5, 1.25 and 2). The extraction was performed at 30 °C in a water bath (Precision Microprocessor Controlled 280, Thermo Scientific, USA) with the required extraction time (30, 60 and 90 min). Then, the tube was centrifuged with a centrifuge (Sorvall T1, Thermo Scientific, USA) at 4000 rpm for 10 min and the supernatant was removed using a pipette and stored in a 4 °C refrigerator for no longer than ten days, until TCC analysis was performed.

The second extraction stage for polyphenols was investigated after the carotenoid extraction optimization conditions were established. Here, polyphenols were extracted from the seaweed residue that remained after carotenoid extraction. For the polyphenol extraction experiments, the SDES composition for optimized carotenoid extraction was also used, and 8% NaOH (w/v) was added to the SDES at a volume ratio of 4:1 to ensure that the SDES was in its hydrophilic phase for these extractions. Thus, the optimization variables investigated for polyphenol extraction were the solvent/solid ratio (v/w), extraction temperature, and extraction time. Prior to each polyphenol extraction experiment, spent seaweed residue from the extraction of carotenoids at optimized conditions, was recovered after centrifugation when the carotenoid extract was separated from the solids. To avoid any hydrophobic SDES remaining in the solids, which would interfere with the solvent-to-solid ratio in the polyphenol extraction, the solids were washed two times with Milli-Q water and then placed in a freeze dryer (FreeZone 2.5 Plus, Labconco, Kansas City, MO, USA) for three days under the operating conditions: freezing temperature at -80 °C, vacuum pressure below 0.13 mbar. After this, the freeze-dried seaweed residue was kept in sealed plastic bags, in a freezer (-20 °C) for no more than a

week before it was used for polyphenol extraction.

Each polyphenol extraction experiment was done in a 50-mL centrifuge tube and involved adding 0.5 g of freeze-dried seaweed residue to an appropriate volume (10, 15 and 20 mL) of hydrophilic SDES at the desired solvent/solid ratio (20, 30 and 40 mL/g). The tubes were then placed in a water bath (Precision Microprocessor Controlled 280, Thermo Scientific, USA) for the desired extraction temperature (40, 50 and 60 °C) and time (30, 50 and 70 min). After this, the tubes were centrifuged at 4000 rpm for 10 min and the hydrophilic SDES was separated from the seaweed residue. Then, HCl solution (2 mol/L) at a volume ratio of 8% (w/v) NaOH (1:1) was added to the hydrophilic SDES for conversion to its hydrophobic phase, resulting in the formation of two layers with the excess aqueous solution. Finally, the centrifuge tubes were centrifuged at 4000 rpm for 10 min, allowing the separation of the liquid layers, where the hydrophobic SDES-rich upper layer was recovered and later used for recycling experiments and the aqueous polyphenol-rich layer was analyzed for TPC and later freeze-dried to obtain polyphenol powder.

2.6. Design of experiments and optimization of TCC and TPC

Box-Behnken design (BBD) was used to study the first and second stage extractions separately. Each experimental design consisted of three input variables with three levels (-1, 0, +1), representing low, medium and high values. The variables and ranges for the experimental design are shown in Tables S1 and S2 (Supplementary material) and were selected based on preliminary single-factor experiments and literature. The experimental design matrices for carotenoid and polyphenol extractions are shown in Tables S3 and S4 (Supplementary material), respectively, where each experimental run was conducted in triplicate.

Response surface methodology was used to optimize the response variables using Design Expert software Stat-Ease 360 (Stat-Ease, Inc., MN, USA) and involved fitting the experimental results to the following nonlinear quadratic model equation:

$$Y^i = \beta_0^i + \beta_1^i X_1^i + \beta_2^i X_2^i + \beta_3^i X_3^i + \beta_{11}^i X_1^{i2} + \beta_{22}^i X_2^{i2} + \beta_{33}^i X_3^{i2} + \beta_{12}^i X_1^i X_2^i + \beta_{13}^i X_1^i X_3^i + \beta_{23}^i X_2^i X_3^i \quad (1)$$

Here, i represents the different extraction stages; Y represents the response variables for the first and second stage extractions where Y^1 is the TCC (mg β -carotene/g dried seaweed) and Y^2 is the TPC (mg gallic acid/g dried seaweed residue), respectively; β_1 , β_2 , and β_3 represent the first-order coefficients; β_{11} , β_{22} , and β_{33} represent the second-order coefficients; β_{12} , β_{13} , and β_{23} represent the interaction coefficients; and X_1 , X_2 , and X_3 represent the coded variables summarized in Tables S1 and S2. To validate the model, extraction experiments were conducted at the conditions predicted for maximizing the TCC and TPC in the first and second stage extractions, respectively. The validation experiments were repeated in triplicate and then compared with the predicted values.

Conventional solvent extractions were also performed to compare SDES extraction performance with commonly used organic solvents. For carotenoids, dried seaweed powder was extracted with hexane under the same extraction conditions used for the optimized hydrophobic SDES extraction (solvent-to-solid ratio 10 mL/g, 30 °C, 90 min), followed by centrifugation (4000 rpm, 10 min) and TCC analysis of the supernatant. For polyphenols, freeze-dried seaweed residue obtained after carotenoid extraction was extracted with 70% ethanol under the same conditions used for the optimized hydrophilic SDES extraction (40 mL/g, 40 °C, 70 min), followed by centrifugation and TPC analysis of the supernatant. All conventional extractions were conducted in triplicate.

2.7. SDES recovery yield after pH-induced two-layer separation

To quantify solvent loss during recovery of the hydrophobic SDES after polyphenol extraction, the recovered hydrophobic SDES-rich upper

layer volume was recorded following pH-induced two-layer separation. Firstly, C10 and C8 fatty acids (C10/C8, molar ratio of 2) were heated at 60 °C in a sealed glass vial to form the optimal hydrophobic SDES. Then, 240 mL of 8% (w/v) NaOH (2 mol/L) was mixed with 60 mL of the liquid to form 300 mL of hydrophilic SDES for TPC extraction. Extraction of TPC from 0.5 g dried seaweed residue occurred under TPC optimal extraction conditions using 20 mL of hydrophilic SDES (40 mL/g of solvent-solid ratio, 40 °C extraction temperature for 70 min). After centrifugation at 4000 rpm for 10 min, the supernatant was transferred to a new 50 mL centrifuge tube. Then 20 mL HCl (2 mol/L) was added to complete the conversion back to hydrophobic SDES and induce separation into two immiscible liquid layers. The hydrophobic SDES-rich upper layer was collected using a pipette and its volume recorded. SDES loss (%) was calculated as:

$$\text{SDES loss (\%)} = \frac{V_0 - V_1}{V_0} \times 100\% \quad (2)$$

Where V_0 is the volume of hydrophobic SDES used to prepare the 20 mL hydrophilic SDES used for the extraction (4.00 mL) and V_1 is the recovered volume of the hydrophobic SDES-rich layer after separation.

2.8. Extraction efficiency of recycled SDES

Hydrophobic SDES was recovered from the second extraction stage after optimal extraction of polyphenols, and then reused to further extract carotenoids from dried seaweed powder. In this study, considering the extraction efficiency of SDES and the stability of carotenoids in SDES, carotenoids were not separated from SDES. Here, seaweed powder (0.5 g) was placed in a 50 mL centrifuge tube and mixed with the recycled SDES under optimal carotenoid extraction conditions. Then, the tube was centrifuged at 4000 rpm for 10 min and hydrophobic SDES separated from the solids and analyzed for TCC. The extraction was also conducted with fresh SDES for comparison and the extraction efficiency (EE) of recycled SDES was calculated according to the following equation:

$$\text{EE(\%)} = \frac{\text{TCC yield in dried seaweed powder using recycled SDES}}{\text{TCC yield in dried seaweed powder using fresh SDES}} \times 100\% \quad (3)$$

2.9. TCC in dried seaweed

Here, the dried seaweed was subjected to repeated extractions with fresh hydrophobic SDES under optimal carotenoid extraction conditions and the TCC was measured after each extraction. These results were used to determine the total TCC in *A. nodosum* by combining the results from six repeated extractions.

2.10. TPC in dried seaweed, carotenoid-extracted residue and recovered hydrophobic SDES

To estimate the total extractable polyphenol content of the dried seaweed and carotenoid-extracted seaweed residue, repeated extractions were performed under optimized polyphenol extraction conditions with hydrophilic SDES prepared as described in Section 2.7. Extractions were repeated until no additional polyphenols were detected in the aqueous polyphenol-rich fraction. The cumulative TPC recovered across extraction cycles was used as an estimate of total extractable TPC in the dried seaweed and carotenoid-extracted residue. In addition, the recovered hydrophobic SDES-rich layer obtained after pH-induced separation was analyzed for residual TPC to assess whether a fraction of phenolics remained associated with the SDES-rich layer.

2.11. Antioxidant activity of bioactive SDES extracts

DPPH radical scavenging activity was used to measure the antioxidant activity of the hydrophobic SDES extract after carotenoid extraction, according to the method used by Zhang et al. (2022a). Additionally, the antioxidant activity of carotenoids in the conventional organic solvent (hexane) was measured to compare with the same concentration of carotenoids in the DES. Briefly, 2, 4, 6, 8 and 10 mL of extracts were taken into five colorimetric tubes, respectively. The volume was fixed to 10 mL by adding SDES or hexane to prepare the sample solutions (20%, 40%, 60%, 80% and 100% v/v). Then, 2 mL of DPPH (0.2 mmol/L) was reacted with 2 mL of sample solutions for 30 min in the dark (Zhang et al., 2022a). The absorbance was determined using a UV-Vis spectrophotometer (Genesys 10S UV/Visible spectrophotometer, Thermo Scientific, USA) at 517 nm to calculate the scavenging activity according to the following equation:

$$\text{DPPH radical scavenging activity (\%)} = \left(1 - \frac{A_1 - A_2}{A_0}\right) \times 100\% \quad (4)$$

Where A_1 : the absorbance of the mixture of 2 mL DPPH and 2 mL sample solution, A_2 : the absorbance of the mixture of 2 mL blank extracted solvent (SDES or hexane) and 2 mL sample solution, A_0 : the absorbance of the mixture of 2 mL blank extracted solvent (SDES or hexane) and 2 mL DPPH.

2.12. Statistical analysis

All results were expressed as the mean $\pm$ standard deviation, with each experiment being carried out in triplicate. The significant variance of preliminary data was analyzed by Minitab Version 18 software (Minitab Inc., PA, USA) using Tukey's multiple range tests at a significance level of $p < 0.05$. Design Expert software Stat-Ease 360 (Stat-Ease, Inc., MN, USA) was used for constructing the experimental design matrices, graph plotting and analyzing the data for optimization.

3. Results and discussion

3.1. Optimization of TCC and TPC

The regression model equation for TCC (Y^1) from the first stage extraction is shown below. The predicted TCC values were calculated according to Eq. (5). The predicted TCC values are summarized in Supplementary material Table S3. Table S3 shows that the experimental values were close to the predicted values, indicating that the regression model accurately predicted TCC values. This is also supported by the ANOVA results in Table 1.

$$\begin{aligned} Y^1 = & 0.9833 - 0.0164\xi_{11} + 0.0900\xi_{12} + 0.1214\xi_{13} - 0.1377\xi_{11}^2 \\ & + 0.0244\xi_{12}^2 + 0.0873\xi_{13}^2 - 0.0100\xi_{11}\xi_{12} + 0.0421\xi_{11}\xi_{13} \\ & - 0.0400\xi_{12}\xi_{13} \end{aligned} \quad (5)$$

Here, ξ_{11} (solvent to solid ratio), ξ_{12} (molar ratio of C10/C8), and ξ_{13} (extraction time) were coded variables, as shown in Supplementary material Table S1.

The regression model equation for TPC (Y^2) from the second stage extraction is shown below. The predicted TPC values were calculated according to Eq. (6). The predicted TPC values are summarized in Supplementary material Table S4. Table S4 showed that the experimental values were close to the predicted values, indicating that the regression model accurately predicted TPC values. This could also be confirmed by ANOVA results in Table 2.

$$\begin{aligned} Y^2 = & 2.7167 + 0.5804\xi_{21} - 0.1588\xi_{22} - 0.2879\xi_{23} - 0.2012\xi_{21}^2 \\ & + 0.0320\xi_{22}^2 + 0.3363\xi_{23}^2 - 0.2375\xi_{21}\xi_{22} + 0.1257\xi_{21}\xi_{23} \\ & - 0.2900\xi_{22}\xi_{23} \end{aligned} \quad (6)$$

Table 1
Analysis of Variance for TCC Extraction.

Source	F-Value	P-Value	Significance
Model	16.15	0.003	**
Linear	28.42	0.001	**
Solvent/Solid Ratio (mL/g)	0.73	0.431	
Molar ratio of C10/C8	31.47	0.002	**
Extraction Time (mins)	40.1	0.001	**
Square	17.12	0.005	
Solvent/Solid Ratio(mL/g) *Solvent/Solid Ratio(mL/g)	28.4	0.003	**
Molar ratio of C10/C8*Molar ratio of C10/C8	0.89	0.388	
Extraction Time (mins)*Extraction Time (mins)	11.4	0.02	*
2-Way Interaction	1.72	0.278	
Solvent/Solid Ratio(mL/g) *Molar ratio of C10/C8	0.19	0.678	
Solvent/Solid Ratio(mL/g) *Extraction Time (mins)	1.86	0.231	
Molar ratio of C10/C8*Extraction Time (mins)	9.87	0.003	**
Error			
Lack-of-Fit	5.97	0.09	
Pure Error			
Total			
S	0.0453767		
R ²	96.67%		
R ² (adj)	90.69%		

* ($p < 0.05$).

** ($p < 0.01$).

Table 2
Analysis of Variance for TPC Extraction.

Source	F-Value	P-Value	Significance
Model	47.52	0.000	***
Linear	97.88	0.000	***
Solvent/Solid Ratio (mL/g)	142.37	0.000	***
Extraction Temperature (°C)	15.22	0.011	*
Extraction Time (mins)	35.02	0.002	**
Square	15.26	0.006	
Solvent/Solid Ratio(mL/g) * Solvent/Solid Ratio(mL/g)	9.42	0.028	*
Extraction Temperature (°C) * Extraction Temperature (°C)	0.24	0.646	
Extraction Time (minutes) * Extraction Time (mins)	26.32	0.004	**
2-Way Interaction	15	0.006	
Solvent/Solid Ratio(mL/g) * Extraction Temperature (°C)	17.03	0.009	**
Solvent/Solid Ratio(mL/g) * Extraction Time (mins)	2.57	0.17	
Extraction Temperature (°C) * Extraction Time (mins)	25.39	0.004	**
Error			
Lack-of-Fit	3.94	0.145	
Pure Error			
Total			
R ²	98.84%		
R ² (adj)	96.75%		

* ($p < 0.05$).

** ($p < 0.01$).

*** ($p < 0.001$).

Here, ξ_{21} (solvent to solid ratio), ξ_{22} (extraction temperature), and ξ_{23} (extraction time) were coded variables as shown in Supplementary material Table S2.

3.2. Analysis of variance for TCC and TPC models

As shown in Table 1, the R² of the model for TCC extraction was 96.67%, indicating that the regression model for the first stage

extraction is a good fit with experimental results (Zhong & Wang, 2010) and the R²(adj) value was 90.69%, meaning the model can explain 90.69% of the response value changes. Moreover, the model's lack of fit ($p > 0.05$) was not significant and the F value of the model was 16.15 ($p < 0.01$), indicating that the model is highly significant. The p-values indicate that the linear terms of X₁₂ (molar ratio of C10/C8) and X₁₃ (extraction time) have a significant impact on the total carotenoid content (TCC) yield, as shown in the contour plots (Fig. 1a, 1b and 1c). With the molar ratio of the C10/C8 increasing, the non-polarity of C10/C8 increases. This corresponds to the principle of “like dissolves like”, which means non-polar compounds like carotenoids are better extracted by non-polar solvents (Viñas-Ospino et al., 2023). Therefore, the TCC yield increases as the molar ratio of C10/C8 increases. Additionally, the interaction term of X₂₃ was also significant, indicating the molar ratio of C10/C8 and extraction time have a synergistic effect on the extraction of TCC, shown in Fig. 1c. A similar trend could also be observed in the study on the optimization of carotenoids from microalgae *Chlorella vulgaris* using deep eutectic solvents by Sportiello et al. (2024).

As shown in Table 2, the ANOVA results also indicated a good fit of the model for TPC in the second stage extraction. The R² for TPC extraction was 98.84%, and the R²(adj) was 96.75%. The lack of fit of the model was insignificant ($p > 0.05$), while the F value of the model was 47.52 ($p < 0.001$), indicating that the model was highly significant. All the linear terms X₂₁ (solvent-to-solid ratio), X₂₂ (extraction temperature), and X₂₃ (extraction time) and the following non-linear terms X₂₁², X₂₃², X₂₁X₂₂ and X₂₂X₂₃ were significant factors for TPC extraction ($p < 0.05$). The contour plots summarizing the interaction effects for TPC extraction are shown in Fig. 2a, 2b and 2c. In a recent study, Zeb et al. (2024) extracted polyphenols from *Saccharina latissima* using deep eutectic solvents composed of betaine and 1,3-butanediol at a molar ratio of 1 to 1. The findings are similar indicating that model terms involving the sample-to-solvents and temperature, as well as the temperature and time both have a significant effect on the yield of total polyphenols (Zeb et al., 2024).

3.3. The validation of the extraction model

The predicted maximum yield of TCC from the model (Eq. (5)) was 1.27 mg β -carotene when 1.0 g of dried *A. nodosum* was extracted using C10/C8 SDES (at a molar ratio of 2:1) at the solvent-to-solid ratio of 10.25 mL/g using an extraction time of 90 min under extraction temperature of 30 °C. To validate the reliability of the extraction model and consider the actual operating conditions, verification experiments were

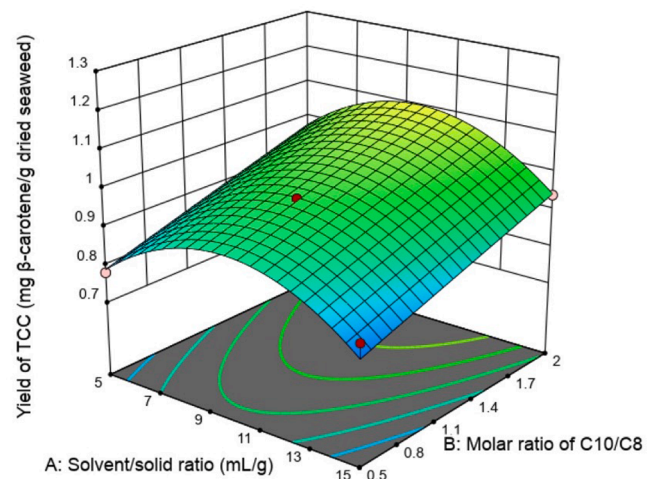


Fig. 1a. Interaction effects of solvent/solid ratio and molar ratio of C10/C8 on yield of total carotenoids.

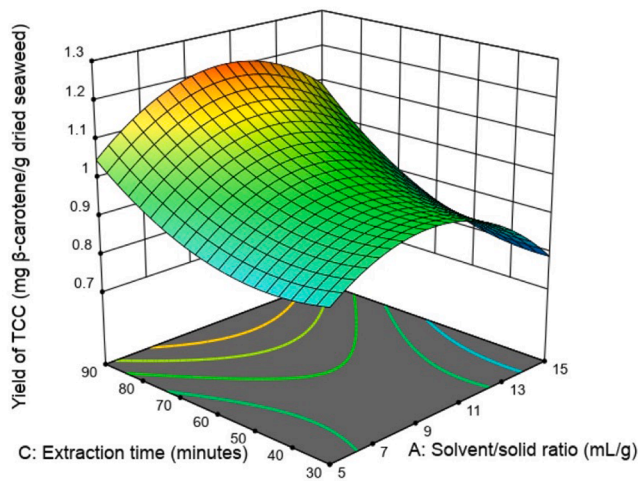


Fig. 1b. Interaction effects of solvent/solid ratio and extraction time on yield of total carotenoids.

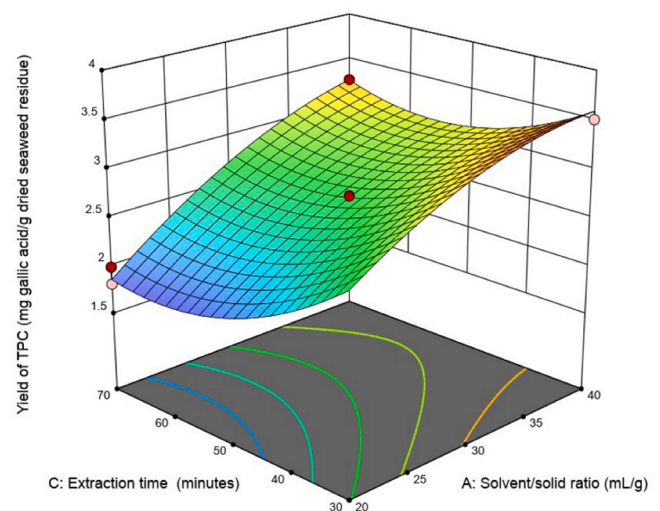


Fig. 2b. Interaction effects of solvent/solid ratio and extraction time on yield of total polyphenols.

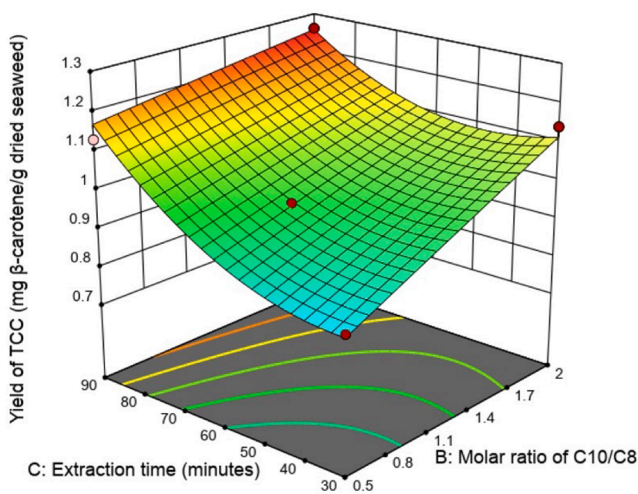


Fig. 1c. Interaction effects of the molar ratio of C10/C8 and extraction time on yield of total carotenoids.

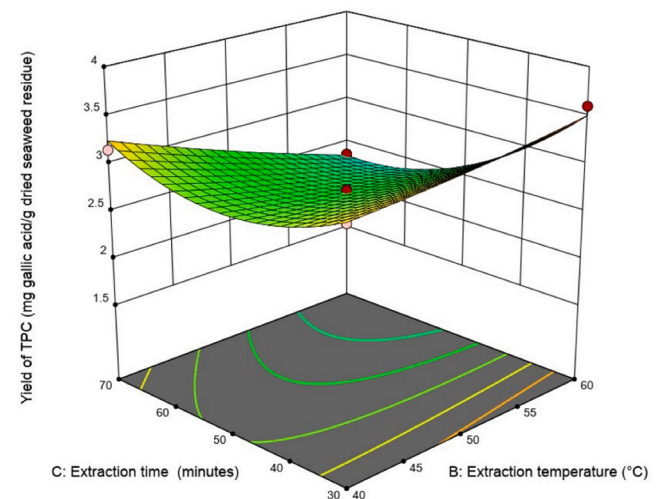


Fig. 2c. Interaction effects of extraction time and extraction temperature on yield of total polyphenols.

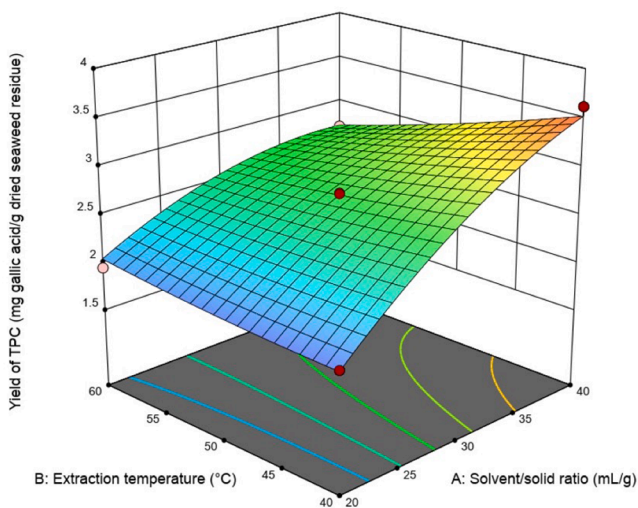


Fig. 2a. Interaction effects of solvent/solid ratio and extraction temperature on yield of total polyphenols.

conducted at the optimal extraction parameters (C10/C8 SDES at a molar ratio of 2:1, solvent-to-solid ratio of 10 mL/g and extraction time of 90 min). The yield of TCC was 1.25 ± 0.02 mg β -carotene/g dried seaweed under the optimal conditions, which was close to the predicted optimal yield (1.27 mg β -carotene/g dried seaweed) and within the predicted 95% confidence interval (1.11, 1.42 mg β -carotene/g dried seaweed). This demonstrated that the extraction model could predict the actual TCC yield well. In comparison, using hexane as the solvent (Garcia-Perez et al., 2022) with the same extraction conditions (solvent-to-solid ratio of 10 mL/g and extraction time of 90 min), the yield of TCC from brown algae was much lower at 0.69 ± 0.01 mg β -carotene/g dried seaweed. This showed that the maximum yield of TCC using the SDES was significantly higher than that of a conventional organic solvent (hexane). In another study, the optimal extraction of carotenoids from *Chlorella vulgaris* using nonanoic acid and dodecanoic acid as a NADES was explored (Sportiello et al., 2024). The results showed that the maximum extraction value was 2.00 ± 0.31 mg/mL under the optimal extraction conditions, which were a 1:1 molar ratio of nonanoic acid to dodecanoic acid, a sample-to-solvent ratio (w/v) of 1:27.33, and an extraction time of 33 min. Thus, the extraction efficiency

of the NADES is influenced by several factors, including the polarity of the mixture, viscosity, solubility in carotenoids, disruption of biomass matrices, and the partition coefficient (Sportiello et al., 2025).

In this study, the predicted maximum yield of TPC was obtained at the solvent-to-solid ratio of 40 mL/g, extraction time of 70 min and extraction temperature of 40 °C. The predicted maximum yield of TPC (Eq. (6)) was 3.99 mg gallic acid when 1.0 g of dried seaweed residue was extracted using hydrophilic phase C10/C8 solvents (at a molar ratio of 2:1). The experimental results conducted under the optimal extraction conditions resulted in a TPC yield of 3.97 ± 0.07 mg gallic acid/g dried seaweed residue, which was close to the predicted optimal yield (3.99 mg gallic acid/g dried seaweed residue) and within the 95% predicted confidence interval (3.525 - 4.451 mg gallic acid/g dried seaweed residue). This demonstrated that the extraction model could predict the actual yield of TPC well. In a previous study, the yield of TPC was 2.54 ± 0.06 mg gallic acid/g dried seaweed residue from a brown seaweed using conventional organic solvent (70% ethanol) (Gunathilake et al., 2024) at the same extraction conditions (the solvent-to-solid ratio of 40 mL/g, extraction time of 70 min and extraction temperature of 40 °C). This revealed that the TPC yield using SDES was significantly greater than with 70% ethanol as the solvent. In comparison, Obluchinskaya et al. (2019) extracted phenolic compounds from *A. nodosum*, comparing 10 NADESs composed of choline chloride, betaine and lactic acid. The results showed that the total phenolic compound values ranged from 7.8 to 50.3 mg/g dried weight (Obluchinskaya et al., 2019). The values were significantly higher than those in this study. This was possibly due to the differences between the hydrophilic SDES and the above NADES as solvents. As well, the pre-treatment of the seaweed could have been a factor, since the *A. nodosum* in this study had been previously solar-dried and this could have resulted in the degradation of polyphenols in the seaweed. Additionally, as part of the sequential extraction process, Milli-Q water was used to wash the seaweed residue three times to remove remaining hydrophobic SDES prior to the extraction of polyphenols, and this could have also removed some of the polyphenols. Furthermore, factors such as seasonal changes and geographical differences can also affect the composition of macroalgae and the yield of the desired compounds in the extracts (Heffernan et al., 2016).

Because the extractions were performed without mechanical stirring, potential mass-transfer limitations should be considered. In the literature, hydrophobic fatty acid-based DES systems have been reported to exhibit low viscosities at or near room temperature (typically ~7–12 mPa·s) (Florindo et al., 2018; Rafique et al., 2026; Suljagić et al., 2026). These values are much lower than common hydrophilic DES such as choline chloride:urea (~1372 mPa·s) (Yadav & Pandey, 2014) but more viscous than hexane (~0.3 mPa·s) (Bolotnikov & Neruchev, 2003). These values indicate that hydrophobic fatty-acid DES are not expected to be sufficiently viscous to impede solvent-biomass contact or mass transfer under the present extraction conditions. However, the absence of agitation may still influence extraction kinetics and future work should examine agitation-assisted extraction and/or quantify viscosity under the specific extraction temperatures used.

3.4. SDES recovery yield after pH-induced two-layer separation

After polyphenol extraction and conversion of hydrophilic SDES back to hydrophobic SDES using HCl, pH-induced two-layer separation was observed. The recovered hydrophobic SDES-rich layer corresponded to a solvent loss of $6.30 \pm 0.84\%$, indicating a SDES recovery yield of $93.70 \pm 0.84\%$ (by volume).

3.5. The extraction efficiency of carotenoids using recovered SDES

The results showed that the extraction efficiency of recycled SDES was 95.20% compared with fresh hydrophobic SDES, which indicated that the recovered hydrophobic SDES can be recycled to extract

carotenoids without a significant decrease in the yield. This has also been demonstrated by other studies. For instance, Zhang et al. (2022a) extracted millet polyphenols using recycled SDES composed of octanoic acid-linalool and found that the yield of millet polyphenols did not decrease significantly when recycled SDES was reused five times (Zhang et al., 2022a). Thus, the recovered SDES has the potential to sustainably maximize the extraction of bioactive compounds from natural products based on its ability to be recycled.

3.6. Total extractable carotenoids in *Ascophyllum nodosum*

The total carotenoids obtained from dried seaweed reached 2 mg β -carotene/g dried seaweed as shown in Table 3, which was achieved after 6 times repeated extraction using fresh fatty acid-based SDES until the solvents were colorless. This aligns with literature values of 0.133–4.43 mg/g dried weight (Čmíková et al., 2024; Walsh et al., 2019) for seaweed. Additionally, it was found that the majority of the carotenoids were extracted after the first two extractions, reaching 1.62 mg β -carotene/g dried seaweed, which accounts for 81% of the total carotenoids present in the dried seaweed.

3.7. Total extractable TPC in *Ascophyllum nodosum*, the carotenoid-extracted residue and distribution between aqueous fraction and recovered SDES

To facilitate interpretation of the sequential extraction process, a summary of carotenoid and polyphenol distribution is presented in Table 4. Repeated hydrophilic SDES extractions performed under the optimized polyphenol extraction conditions indicated that the total extractable TPC in the original as-received *A. nodosum* was 10.04 mg GAE/g dried seaweed, determined after eight extraction cycles, with no additional polyphenols detected after the sixth cycle.

Following carotenoid extraction, the total extractable TPC in the carotenoid-extracted residue was 7.08 mg GAE/g dried residue, indicating that approximately 70.5% of the initial extractable polyphenols remained in the residue. This demonstrates that the hydrophobic SDES extraction step selectively targets carotenoids while largely preserving hydrophilic phenolic compounds for subsequent recovery.

The first two extraction cycles recovered the majority of polyphenols from the residue (3.93 ± 0.06 and 1.76 ± 0.06 mg GAE/g dried residue, respectively), corresponding to approximately 80.4% of the total extractable TPC in the residue. Relative to the original seaweed biomass, the first extraction step recovered approximately 39% of the total extractable polyphenols, while the optimized single-step extraction recovered approximately 56% of the extractable polyphenols present in the residue.

In this work, the aqueous fraction was treated as the primary polyphenol-rich fraction. However, repeated extraction experiments demonstrated that a small fraction of phenolics remained associated with the recovered hydrophobic SDES-rich layer (0.18 mg GAE/g dried residue total; ~2.5% of total extractable residue TPC), occurring only in the first two cycles. This suggests that a minor subset of more hydrophobic phenolic constituents may preferentially associate with the

Table 3

The carotenoid yield from dried seaweed powder after repeated extraction with fresh SDES under optimal extraction conditions.

Number of extractions	1	2	3	4	5	6	Total amount of carotenoids
Carotenoid yield [#] (β -carotene/g dried seaweed)	1.25	0.37	0.17	0.10	0.07	0.04	2.00

[#] Results are shown for the last extraction.

Table 4

Summary of extractable carotenoids and polyphenols and their distribution during sequential extraction.

Component	Value (mg/g)	% of total	Notes
Total carotenoids (original biomass)	2.00	100%	Determined by repeated extraction
Carotenoids (first two cycles)	1.62	81%	Rapid extraction phase
Total polyphenols (original biomass)	10.04	100%	Determined by repeated extraction
Polyphenols in residue	7.08	70.5%	Retained after carotenoid extraction
Polyphenols (1st extraction)	3.93	39.1%	Relative to original biomass
Polyphenols in SDES layer	0.18	2.5%	Minor retained fraction

SDES-rich layer. Importantly, the magnitude of this retained fraction was small, indicating that the switching/separation approach effectively recovered the vast majority of polyphenols into the aqueous fraction under the applied conditions.

Comparison of the total extractable TPC in the original seaweed and the carotenoid-extracted residue provides insight into polyphenol distribution during the sequential process. Although the majority of polyphenols were retained in the residue, a reduction relative to the original biomass was observed. This reduction can be attributed to both processing losses and extraction-related distribution effects. Experimental observations indicated that approximately 17.61% of polyphenols were lost during water washing, rotary evaporation and freeze-drying, while additional differences may arise from redistribution or incomplete recovery during processing.

It is important to note that the polyphenol extraction step is integral to the sequential biorefinery concept of this study. Following carotenoid extraction, the remaining seaweed residue still contains a substantial fraction of hydrophilic bioactives. Recovering this additional fraction increases overall biomass valorization and reduces waste. Moreover, pH-induced two-layer separation enables recovery of a hydrophobic SDES-rich layer suitable for reuse in carotenoid extraction, linking the second step directly to solvent recyclability and process sustainability rather than serving solely as a demonstration of switchability.

3.8. DPPH radical scavenging activity

As shown in Fig. 3, the DPPH radical scavenging activity of carotenoids using hexane as a solvent was higher than that of carotenoids in SDES at the same carotenoid concentration. Literature reports indicate

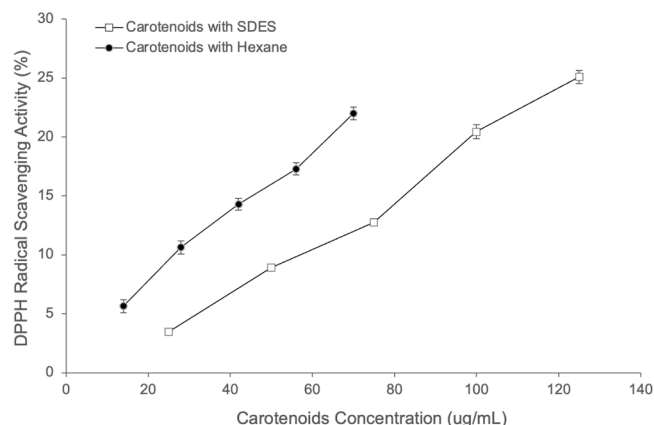


Fig. 3. DPPH radical scavenging activity of carotenoids in SDES and hexane at equivalent carotenoid concentrations. Error bars represent mean $\pm$ standard deviation ($n = 3$).

that DPPH radical scavenging outcomes depend strongly on solvent polarity (Dawidowicz et al., 2012; La et al., 2021; Yamauchi et al., 2016), and that carotenoid antioxidant behavior is influenced by the surrounding solvent environment (Al Tarifi et al., 2020; Bulut et al., 2019; Gao et al., 2020). For example, La et al. (2021) demonstrated substantial changes in DPPH reactivity across solvents at the same antioxidant concentration, and Yamauchi et al. (2016) reported that DPPH values varied markedly between solvent systems, highlighting the role of the assay medium.

The lower DPPH radical scavenging activity measured for carotenoids in SDES should therefore not be interpreted as a direct loss of carotenoid bioactivity. Rather, it may reflect solvent-dependent assay behaviour and/or interactions between carotenoids and the fatty acid-based SDES that reduce the apparent radical scavenging activity under the DPPH assay conditions. Analogous “masking” phenomena have been reported when molecular interactions reduce the apparent electron donating ability of antioxidants (e.g., protein–flavonoid binding effects; Geng et al., 2019).

The reduced DPPH radical scavenging activity observed in SDES is therefore likely influenced by solvent-dependent effects, and further investigation is needed to determine whether these effects are reversible or involve partial degradation. This may include evaluating antioxidant activity after dilution, solvent exchange, release from a food or packaging matrix, or separation of carotenoids from the SDES.

Such studies would clarify whether SDES can function as a temporary carrier during extraction and processing while still allowing carotenoids to exert antioxidant activity after release or formulation. In packaging or food matrices, dilution and/or release may reduce SDES–carotenoid association, potentially allowing antioxidant function to be expressed, and this should be evaluated in future formulation studies.

4. Conclusions and future work

This study demonstrates the feasibility of using a pH-responsive switchable deep eutectic solvent (SDES) system for the sequential extraction of carotenoids and polyphenols from *Ascophyllum nodosum*. The fatty acid-based SDES enabled polarity switching and pH-induced two-layer separation, allowing efficient recovery of both hydrophobic and hydrophilic bioactive compounds within a single process. The optimized SDES achieved carotenoid and polyphenol yields of 1.25 mg β -carotene equivalents/g and 3.97 mg GAE/g dried seaweed, respectively, exceeding those obtained using conventional solvents. Repeated extraction experiments showed that the total extractable carotenoid content was 2.00 mg/g, with approximately 81% recovered within the first two extraction cycles, indicating efficient extraction. The total extractable polyphenol content of the original seaweed was 10.04 mg GAE/g, of which approximately 70.5% remained in the residue after carotenoid extraction, confirming selective extraction of carotenoids while preserving hydrophilic phenolics. A single extraction step recovered approximately 56% of the extractable polyphenols in the residue (~39% of the total), and only a minor fraction (~2.5%) remained associated with the SDES-rich layer. The SDES exhibited strong recyclability, with a recovery yield of 93.70% and retention of 95.20% carotenoid extraction efficiency after reuse. The reduced antioxidant activity observed in SDES is likely influenced by solvent-dependent effects, and further investigation is needed to determine whether these effects are reversible or involve partial degradation.

Overall, these findings suggest that fatty acid-based SDES can support a sustainable biorefinery-oriented approach for valorizing *A. nodosum* by enabling sequential recovery of multiple bioactive fractions and facilitating solvent reuse. Future work should investigate: application to other seaweed species, scale-up considerations and process mass balances (including SDES losses during recovery), formulation and release behavior of SDES–carotenoid extracts in real food/packaging matrices, and economic and life-cycle assessment of SDES recovery relative to conventional solvent extraction.

Ethical statement

No humans or animals were involved in this study.

Funding

This work was funded by a Discovery Grant (RGPIN-2020-04397) from the Natural Sciences and Engineering Research Council of Canada (NSERC).

Code availability

Not applicable.

CRediT authorship contribution statement

Desheng Ji: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marianne Su-Ling Brooks:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.afres.2026.102392](https://doi.org/10.1016/j.afres.2026.102392).

Data availability

Data will be made available on request.

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